

Distributional Treatment Effects and Policy Spillovers

Dissertation proposal: four chapters

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Policy effects can differ across the outcome distribution and can reach people outside the directly targeted population.

Two linked questions

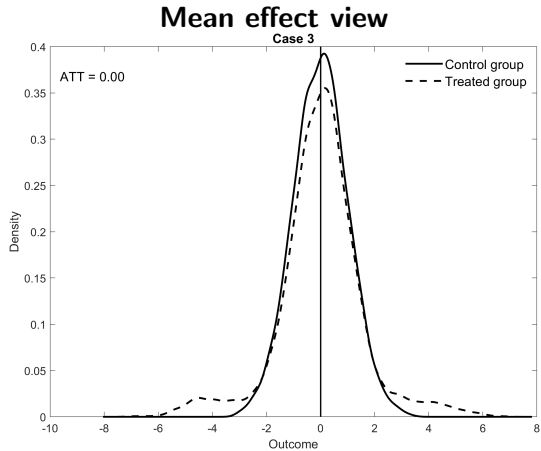
How should we estimate and diagnose distributional treatment effects? What can a local policy change outside its immediate target population?

Chapters 1–2 develop the distributional methods agenda. Chapters 3–4 apply policy evaluation to immigration enforcement and opioid-settlement spending.

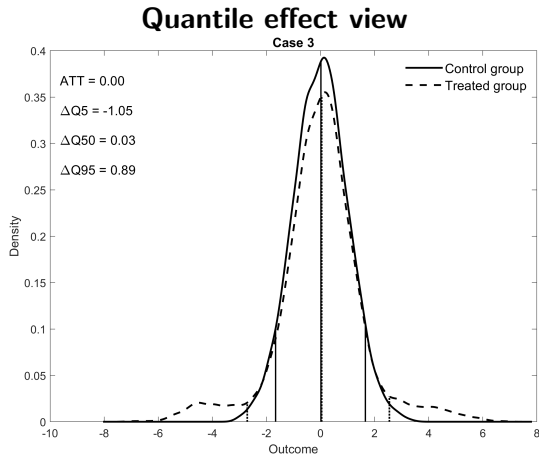
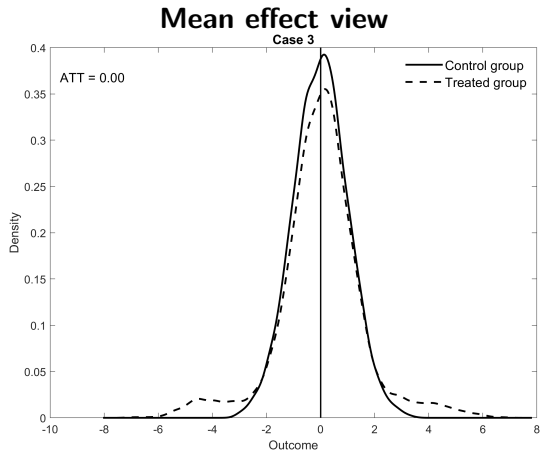
Roadmap

- ① Chapter 1: survey of distributional DiD and QTT estimators.
- ② Chapter 2: a diagnostic for distributional non-divergence and the ATT/QT decision.
- ③ Chapter 3: sanctuary policies, ICE cooperation, and Medicare spillovers.
- ④ Chapter 4: county opioid-settlement spending and overdose mortality.
- ⑤ Synthesis and questions.

A mean can hide the policy-relevant story



A mean can hide the policy-relevant story



There is no average effect while some parts of the distribution gain and some lose.

When do we need QTT in addition to ATT

Pattern	Example	What happened
Winner-takes-all	Small class assignment in Project STAR: Jackson and Page (2013)	Small classes improved achievement, but gains were largest near the top of the achievement distribution. Mean effects miss that high-achieving students gained more.
Opposite signs	Teach For America effects on reading: Penner (2016)	TFA had negative effects at the bottom of the reading distribution but positive effects in the upper half. The average can hide winners and losers.
Tail-only effects	Colombian school vouchers: Lamarche (2011)	Voucher effects were strongest in the lower tail, especially among weaker students. This is a clean “vulnerable students move most” example.
Dispersion effect	Health insurance reference pricing: Aouad, Brown, and Whaley (2021)	Reference pricing changed the distribution of medical spending, with large effects in the right tail of insurer spending. The policy affected risk and dispersion, not just the mean.

DTT: distributional effect

$$\text{DTT}(y) = F_{Y_{11}^I}(y) - F_{Y_{11}^N}(y).$$

How did treatment change the probability of being below threshold y ?

QTT: quantile effect

$$\text{QTT}(\alpha) = F_{Y_{11}^I}^{-1}(\alpha) - F_{Y_{11}^N}^{-1}(\alpha).$$

How did treatment shift the α th quantile of treated outcomes?

The problem: missing untreated counterfactual distribution: $F_{Y_{11}^N}(y)$.

Once we have this CDF

We can compute both the distributional effect and the quantile effect.

Chapter 1: one missing distribution, several answers

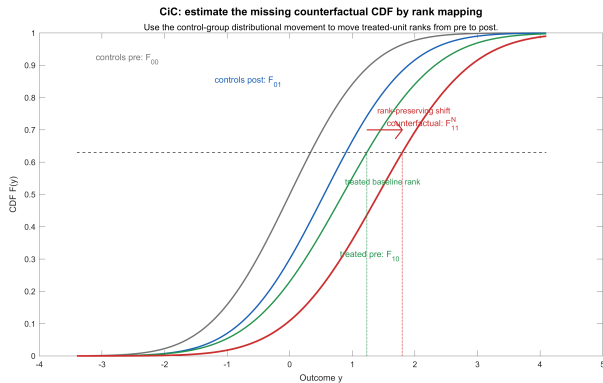
$$F_{11}^0(y) = \Pr(Y_1(0) \leq y \mid G = 1)$$

- We observe the treated group after treatment, but never its untreated outcome distribution at that date.
- Each QTT method supplies a different rule for constructing this same missing CDF.
- The data structure and the economic argument determine which rule is credible.

Interpretation

QTT compares two *marginal quantiles*; it is generally not the quantile of individual treatment gains.

Changes-in-changes: transport ranks



$$Q_{11}^0(\tau) = Q_{01}(F_{00}(Q_{10}(\tau))) .$$

What identifies the CiC counterfactual?

- 1 Untreated outcomes are an increasing function of a scalar latent rank.
- 2 The latent-rank distribution within each group does not change over time.
- 3 Treated baseline ranks lie within the comparison group's support.

What to inspect

Baseline support, changes in composition, and placebo rank transports using earlier untreated periods.

CiC need not satisfy parallel changes at fixed thresholds.

CiC with covariates: transport, then aggregate

- 1 Within $X = x$, apply the control rank map to treated baseline outcomes.
- 2 Obtain a conditional counterfactual CDF $F_{11}^0(y | x)$.
- 3 Average these CDFs over the treated-post covariate distribution.

$$F_{11}^0(y) = \int F_{11}^0(y | x) dF_{X|G=1, T=1}(x).$$

Additional burden

Conditional rank stability and overlap in X ; average conditional CDFs, not conditional quantiles.

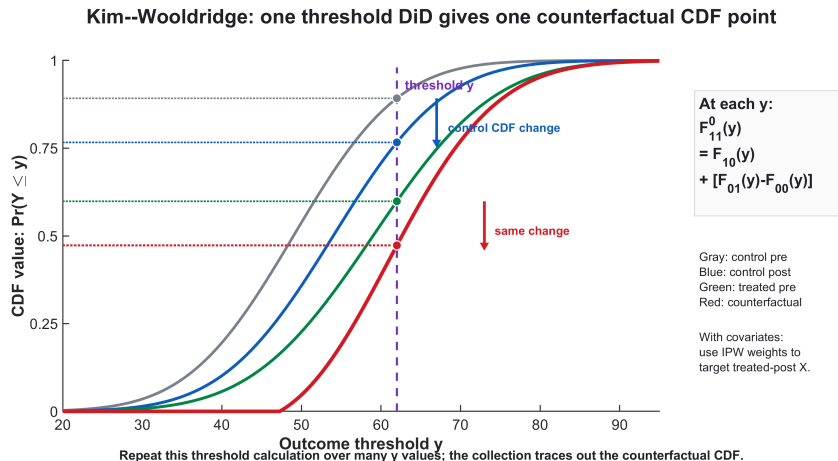
Distributional parallel trends: compare thresholds

$$F_{11}^0(y) = F_{10}(y) + F_{01}(y) - F_{00}(y).$$

- At each fixed outcome threshold y , add the control group's CDF change to the treated baseline CDF.
- Check pretreatment threshold trends and whether the implied counterfactual is a valid, nondecreasing CDF.
- This restriction is distinct from parallel trends in means or in quantiles.

A parallel mean trend can hold when threshold trends fail.

Kim–Wooldridge: a DiD at each threshold



Repeating the calculation over y traces the counterfactual CDF.

Kim–Wooldridge: reweight to treated-post X

Let $p_{gt}(X) = \Pr(G = g, T = t \mid X)$ and $p_{11} = \Pr(G = 1, T = 1)$.

$$F_{11}^0(y) = \mathbb{E} \left[\frac{p_{11}(X)}{p_{11}} \left\{ \frac{1_{10}}{p_{10}(X)} + \frac{1_{01}}{p_{01}(X)} - \frac{1_{00}}{p_{00}(X)} \right\} 1\{Y \leq y\} \right].$$

- The three observed untreated cells supply the threshold-DiD contrast.
- Weights make their covariate mix comparable to treated units after treatment.
- Identification needs conditional threshold trends and overlap in all cells.

Transitions and threshold methods use different data

Callaway–Li panel QTT

- Linked units reveal the distribution of untreated changes.
- Transport those changes to treated units using a stated restriction on baseline–change dependence.
- Covariates can align the comparison population.

Kim–Wooldridge distributional DiD

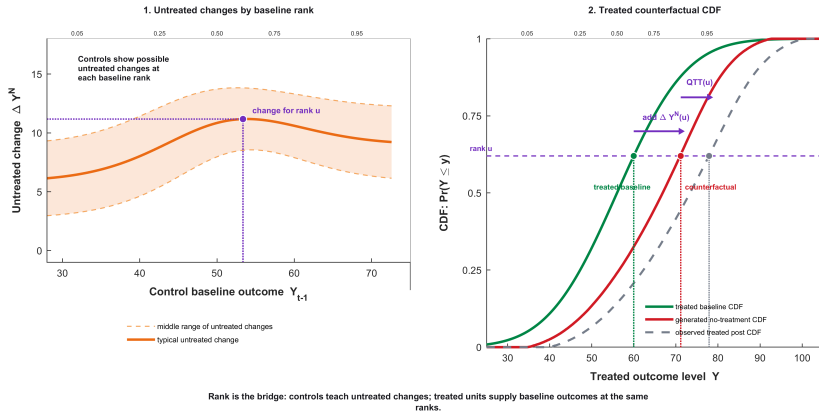
- Repeated cross-sections can identify CDF changes at fixed thresholds.
- With covariates, reweight comparison cells to the treated-post covariate distribution.
- Overlap and conditional threshold trends are required.

What panel data add

The joint movement of baseline levels and subsequent changes is observed, but its untreated stability still requires an assumption.

Callaway-Li: learn untreated changes from linked units

Callaway-Li: learn changes by rank, then build the counterfactual CDF



The shaded range represents possible changes at a baseline rank; one curve is only a teaching shorthand.

Callaway–Li: from changes to one CDF

$$Y_t(0) = Y_{t-1}(0) + \Delta Y_t(0).$$

- 1 Estimate the distribution of untreated changes from panel transitions.
- 2 Specify how those changes depend on the treated group's baseline outcomes.
- 3 Combine baseline and change for each treated unit, then pool the constructed outcomes into one counterfactual CDF.

Identification

The marginal change distribution alone is insufficient. The baseline–change dependence (copula) matters, especially in the tails.

Panel QTT: what should be checked?

- Compare untreated-change distributions over available pretreatment transitions.
- Plot baseline-change association and rank mobility in treated and comparison groups.
- Inspect attrition and support, especially at the tails.
- Show sensitivity to alternative dependence assumptions when the coupling is hard to defend.

Linked records reveal transitions, but cannot reveal treated units' missing untreated transitions.

Staggered adoption changes the target

- With several adoption dates, estimate an untreated counterfactual distribution for each cohort g and time $t \geq g$.
- Use never-treated or not-yet-treated comparisons and an explicit cohort-time identifying restriction.
- Report $QTT_{g,t}(\tau)$ before aggregating; weights determine the pooled target.

$$QTT_{g,t}(\tau) = Q_{Y_t(1)|G=g}(\tau) - Q_{Y_t(0)|G=g}(\tau).$$

A single pooled curve can hide cohort and event-time heterogeneity.

Staggered QTT: comparison and aggregation choices

- For cohort g , build the untreated CDF at each $t \geq g$ from never-treated or not-yet-treated units.
- State the cohort-specific restriction: threshold trends, rank transport, or a panel dependence model.
- Aggregate group-time QTTs only after choosing weights and a common event-time support.

$$\theta_e(\tau) = \sum_{g: g+e \leq T} w_{g,e} QTT_{g,g+e}(\tau), \quad \sum_g w_{g,e} = 1.$$

Changing cohort weights can change the pooled curve.

Chapter 1: method choice in one view

Method	Main data	Restriction constructing the missing CDF
Changes-in-changes	Panel or repeated cross-section	Common monotone rank map
Threshold DiD	Panel or repeated cross-section	Parallel CDF changes at fixed thresholds
Panel QTT	Linked panel	Untreated-change distribution and dependence stability
Conditional/IPW	Covariates and overlap	A base distributional restriction conditional on X
Staggered QTT	Multiple cohorts and periods	Cohort-time restriction plus comparison-set choice

Better prediction of an observed CDF does not establish the missing CDF.

Inference must rebuild the entire counterfactual

- 1 Resample at the policy assignment level (states in the minimum-wage illustration).
- 2 Re-estimate CDFs, weights, rank maps, or dependence parameters in each draw.
- 3 Reconstruct and invert the counterfactual CDF in each draw.
- 4 Report a simultaneous band when discussing a curve across many quantiles.

Two kinds of uncertainty

Sampling bands condition on the identifying restriction. Comparing plausible restrictions or reporting bounds addresses uncertainty about identification.

When dependence is not credible: retain bounds

- The baseline distribution and untreated-change distribution do not by themselves determine the distribution of their sum.
- Bounds allow all baseline–change couplings consistent with the maintained marginals.
- A wide identified set is a substantive result: the data do not choose a unique QTT.

A precise estimate under one copula can coexist with a wide dependence-robust set.

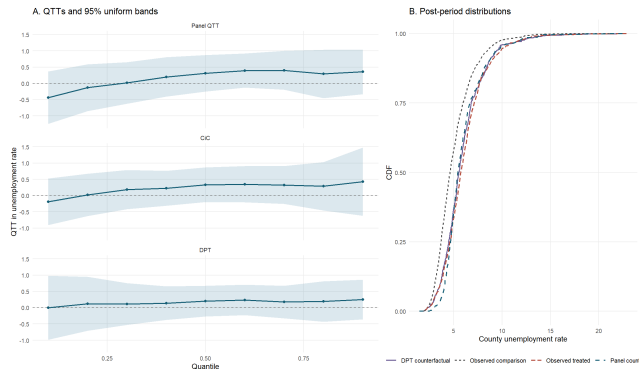
Chapter 1 simulation: assumptions determine bias

True restriction in the design	CiC bias	Threshold-DiD bias
CiC only	0.001	0.122
Threshold DiD only	0.311	0.002
Neither	0.269	0.376

- Values are from the survey's assumption laboratory at 4,000 observations per cell.
- Larger samples reduce sampling noise, not bias from a false identifying restriction.

Chapter 1 illustration: minimum wages

- County unemployment, 2005–2007; 678 treated counties and 1,682 comparisons.
- Panel QTT, CiC, and threshold DiD use the same county target but give different lower-tail estimates.
- State-clustered uniform bands include zero; dependence-robust bounds remain wide.



Agreement across estimators is not identification.

Minimum-wage illustration: assumption sensitivity

QTT estimate (unemployment percentage points)	$\tau = 0.10$	$\tau = 0.50$
Panel QTT	-0.437	0.307
Changes-in-changes	-0.188	0.333
Threshold DiD	-0.001	0.201
Dependence-robust identified set	$[-1.516, 1.267]$	$[-1.008, 1.409]$

- All point estimators use the same treated-county target but different counterfactual restrictions.
- Uniform state-clustered bands include zero across the reported quantiles.
- The bounds include zero and effects of both signs at these quantiles.
- These bounds still maintain a transferable untreated-change distribution.

Minimum-wage illustration: what diagnostics say

- Pretreatment placebo QTT processes do not reject, but their simultaneous bands are wide.
- The baseline–change association shifts for eventually treated counties; the post-policy shift is not itself a test of their missing untreated copula.
- Only 11 treated and 20 comparison states contribute independent assignment clusters.
- Leave-one-state-out ranges show influence on magnitudes, especially in the tails.

Decision

The exercise illustrates conditional answers under different restrictions; it does not establish a preferred minimum-wage effect.

Chapter 1 lesson and the question for Chapter 2

- ① Define the distributional target and the treated population.
- ② State the rule that constructs the missing untreated CDF.
- ③ Diagnose observable implications and report uncertainty at the policy assignment level.
- ④ Use bounds or sensitivity analysis when no point restriction is persuasive.

Next question

Before choosing a QTT estimator, can we detect instability in the comparison distribution and decide whether the ATT misses a material distributional pattern?

From estimators to diagnostics

Chapter 1's methods share a sequence:

- 1 Choose a comparison group and a counterfactual restriction.
- 2 Construct the missing untreated CDF.
- 3 Estimate DTT or QTT and its uncertainty.

The diagnostic gap

A credible comparison distribution is central to the argument, but mean pretrends alone do not assess its distributional stability.

Chapter 2 question

Can we detect comparison-group instability and quantile separation before selecting a QTT estimator?

Chapter 2: what the paper contributes

Joint work with Donggyu Sul.

- 1 Defines distributional σ -convergence: stability around quantiles, not only around the mean.
- 2 Uses slope tests to detect distributional divergence.
- 3 Applies a sieving algorithm if the groups are unstable.
- 4 Classifies whether the post-treatment evidence supports ATT or requires QTT.

Important

The diagnostic flags where direct QTT estimation is needed. It does not identify the counterfactual CDF or establish its causal assumptions.

The idea of σ -convergence: Kong, Phillips, and Sul (2019)

A group, $G \in \{N, I\}$, is weak σ non-diverging if its cross-sectional dispersion is not growing over time.

Weak σ statistic

$$K_{G,t} = \frac{1}{n_G} \sum_i (y_{it}^G - \mu_{G,t})^2.$$

What this mean

Are units in the group spreading farther away from their own mean?

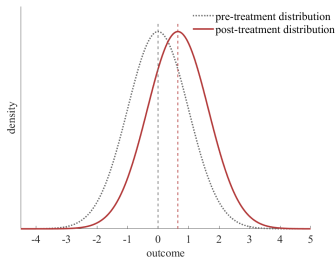
If yes, the group may not share a common benchmark trend.

The idea of σ -convergence

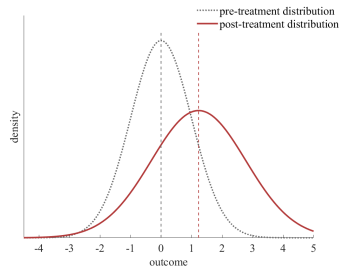
Weak σ statistic

$$K_{G,t} = \frac{1}{n_G} \sum_i (y_{it}^G - \mu_{G,t})^2.$$

Non-diverging



Diverging



Why weak σ is not enough for QTT

Weak σ -convergence asks whether dispersion around the mean grows.

But a distribution can change shape even if mean-centered dispersion looks stable:

- the lower tail may move down,
- the upper tail may move up,
- changes may offset around the mean.

For QTT

We need to know whether the distribution is stable around relevant quantiles.

Distributional σ -convergence

For quantile α , define quantile-centered dispersion for a group, $G \in \{N, I\}$:

$$K_{G,t}^{\alpha} = \frac{1}{n_G} \sum_i (y_{it}^G - \mu_{G,t}^{\alpha})^2.$$

Definition

Group G is distributionally non-diverging if $K_{G,t}^{\alpha}$ has no positive trend for relevant quantiles α .

Intuition

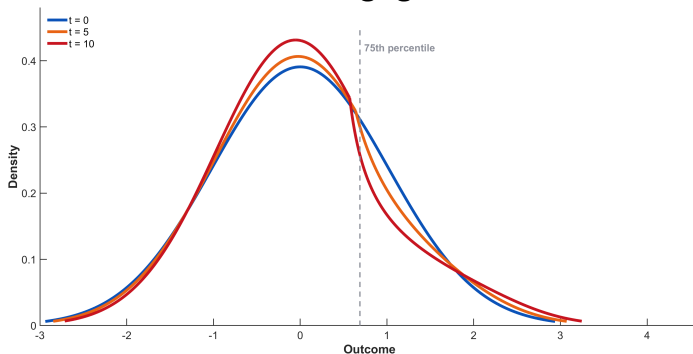
Instead of asking only whether units move away from the mean, ask whether they move away from the lower tail, median, and upper tail.

Distributional σ -convergence

Weak σ statistic

$$K_{G,t}^{\alpha} = \frac{1}{n_G} \sum_i (y_{it}^G - \mu_{G,t}^{\alpha})^2.$$

Diverging



Treated control quantile separation

For quantile α , define treated control quantile separation statistics:

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \frac{1}{n_N} \sum_i (y_{it}^N - \hat{\mu}_{I,t}^\alpha)^2.$$

Definition

Treated group is distributionally non-diverging from control group if $K_{N,t}^{\Delta(\alpha)}$ has no positive trend for relevant quantiles α .

Intuition

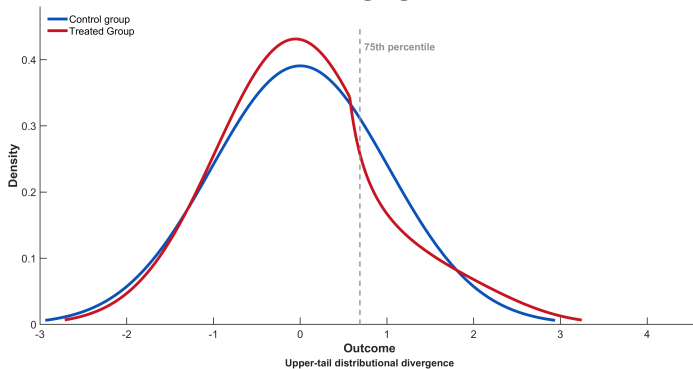
Ask whether the relevant quantiles α of treated group move away from the control group distribution.

Treated control quantile separation

Weak σ statistic

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \frac{1}{n_N} \sum_i (y_{it}^N - \hat{\mu}_{I,t}^\alpha)^2.$$

Diverging



Three diagnostic objects summary

Object	What it measures	Why it matters
$\hat{K}_{G,t}$	Dispersion around group mean	Weak stability of a group
$\hat{K}_{G,t}^{\alpha}$	Dispersion around group quantile	Distributional stability of a group
$\hat{K}_{N,t}^{\Delta(\alpha)}$	Distance from controls to treated quantile	Treated-control quantile separation

Each object is converted into a simple time-series trend test.

The slope test

For each diagnostic object, run

$$\hat{K}_{m,t} = \phi_{m,0} + \phi_m t + u_{m,t}.$$

Decision rule

$$t_{\hat{\phi}_m} > 1.65 \quad \Rightarrow \quad \text{positive divergence.}$$

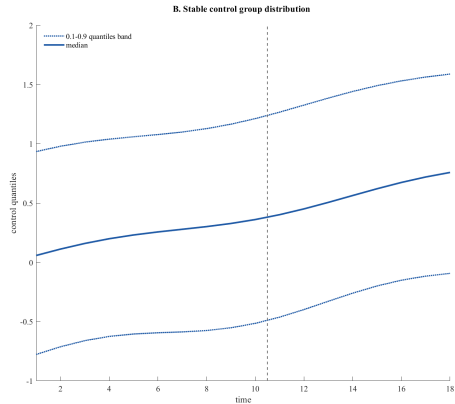
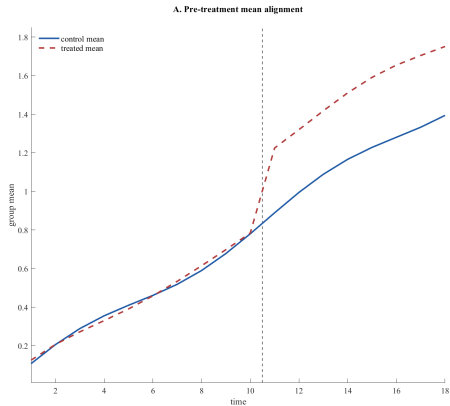
What this means

A positive trend means the group, quantile, or treated-control distance is spreading out over time.

Assumptions in plain language

Label	Name	Role
A1	Mean pre-condition	Treated and control means are comparable before treatment.
A2	Non-diverging control distribution	Controls can serve as a stable distributional benchmark; if not, sieve.
A3	Non-diverging treated distribution before treatment	Treatment group can serve as a stable distributional benchmark before treatment; if not, sieve.
A4	Pre-treatment distributional comparability	Treated-control quantile gaps are not already diverging.
A5	Constant average effect when ATT is nonzero	A nonzero ATT can be read as a stable location shift.
A6	Bounded remainder under divergence	Technical condition that lets the trend dominate noise.

Assumptions A1-A2



Step 1: validate the control benchmark

The control group must be stable over the period in which it is used as a counterfactual benchmark.

$$\hat{K}_{N,t} \quad \text{and} \quad \hat{K}_{N,t}^{\alpha}$$

should not have positive trends.

Why this matters

If controls diverge after treatment begins, treated-control quantile differences may reflect control instability rather than treatment effects.

Step 2: sieve if the control group diverges

Raw controls

- May include units with different trends or shocks.

Sieve

- Remove divergent units.
- Re-test retained units.

Retained controls

- Used only if non-diverging.

Interpretation

Sieving is a research-design step: construct a credible comparison distribution before asking about QTT.

*Sieving technique is modified from Phillips and Sul (2007), Lee et al. (2026) and Lee and Sul (2026)

Step 3: validate the treated group before treatment

The treated group must be stable before treatment starts.

$$\hat{K}_{l,t} \quad \text{and} \quad \hat{K}_{l,t}^{\alpha}$$

should not have positive trends.

Why this matters

If treated group diverge before treatment, the comparison between treated and control group isn't valid.

Step 4: check pre-treatment comparability

Even if each group is internally stable, treated and control distributions may already be drifting apart before treatment.

Cross-quantile distance

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \frac{1}{n_N} \sum_i (y_{it}^N - \hat{\mu}_{I,t}^{\alpha})^2.$$

Purpose

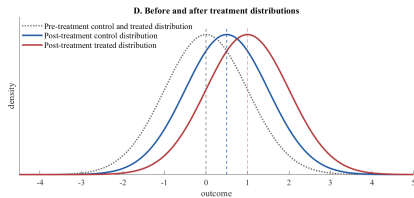
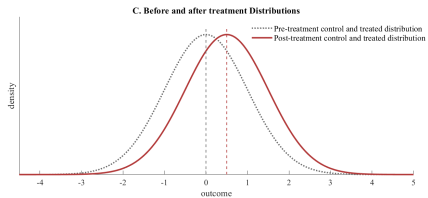
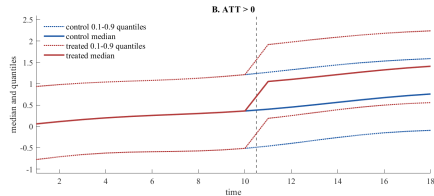
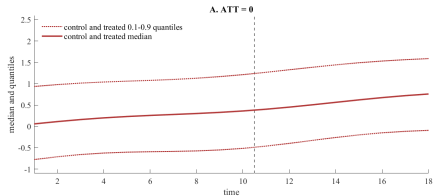
Rule out pre-existing quantile divergence before interpreting post-treatment separation as policy evidence.

Step 5: classify the post-treatment pattern

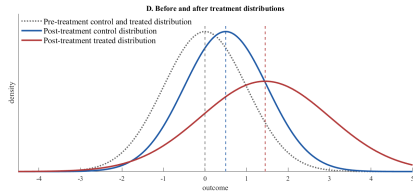
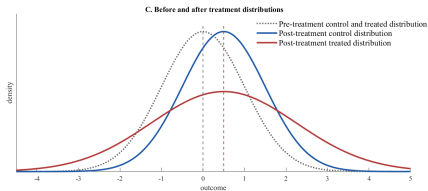
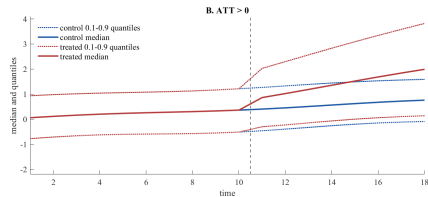
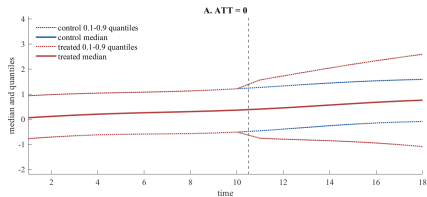
Once the control group is distributionally stable for all time period, treated group is distributionally stable before treatment, and the cross-quantile distance is stable before treatment, ask whether the treated group changes in a way ATT can summarize.

Case	Diagnostic pattern	Empirical message
(i)	Treated distribution non-diverging	ATT is enough: no QTT if ATT is zero; common shift if ATT is stable nonzero.
(ii)	Treated distribution diverges, ATT near zero	Mean says little, but distribution moves; estimate QTT.
(iii)	Treated distribution diverges, ATT nonzero	ATT exists but is incomplete; estimate QTT.

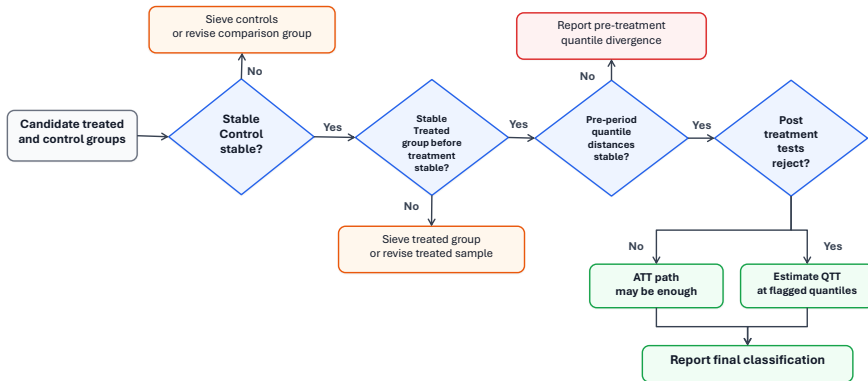
Case (i)



Case (ii) and Case (iii)



Decision tree for applied researchers



Theory intuition: within-group divergence

The quantile-centered statistic can be written as

$$\hat{K}_{G,t}^{\alpha} = \hat{K}_{G,t} + (\hat{d}_{G,t}^{\alpha})^2, \quad \hat{d}_{G,t}^{\alpha} = \hat{\mu}_{G,t} - \hat{\mu}_{G,t}^{\alpha}.$$

Key idea

If the distance between the mean and a quantile grows over time, then the statistic inherits a positive trend.

What it means

A tail moving away from the center shows up as an upward trend in quantile-centered dispersion.

Within-group result

Stable mean-to-quantile distance

$$\hat{d}_{G,t}^{\alpha} = c_{\alpha} + O_p(n_G^{-1/2}) \Rightarrow t_{\hat{\phi}_{G,\alpha}} \Rightarrow N(0, 1).$$

Growing mean-to-quantile distance

$$\hat{d}_{G,t}^{\alpha} = c_{\alpha} t^{\gamma} + O_p(1), \gamma > 0 \Rightarrow t_{\hat{\phi}_{G,\alpha}} \rightarrow +\infty.$$

The test separates bounded distributional movement from growing distributional movement.

Theory intuition: cross-quantile separation

For cross-group diagnostics,

$$\hat{\Delta}_t^\alpha = \hat{\mu}_{I,t}^\alpha - \hat{\mu}_{N,t}^\alpha$$

is the treated-control quantile gap.

Main decomposition

$$\hat{K}_{N,t}^{\Delta(\alpha)} = \hat{K}_{N,t}^\alpha + (\hat{\Delta}_t^\alpha)^2 - 2\hat{\Delta}_t^\alpha e_t.$$

Interpretation

If controls are stable, an upward trend comes from treated quantiles separating from controls.

Cross-group result

Stable quantile gap

$$\hat{\Delta}_t^\alpha = c_\alpha^\Delta + O_p(n_{\min}^{-1/2}) \Rightarrow t_{\hat{\phi}_{N,\Delta,\alpha}} \Rightarrow N(0, 1).$$

Growing quantile gap

$$\hat{\Delta}_t^\alpha = c_\alpha^\Delta t^\gamma + O_p(1), \gamma > 0 \Rightarrow t_{\hat{\phi}_{N,\Delta,\alpha}} \rightarrow +\infty.$$

Important for interpretation

The statistic is based on squared distances. It detects separation, not whether QTT is positive or negative.

Monte Carlo design in words

The simulations create panel data where the true treatment pattern is known. They check whether the diagnostic separates common shifts from distributional change.

Case (i)

- Treatment is a pure location shift.
- ATT is a sufficient summary.
- Diagnostic should not call for QTT.

Cases (ii)–(iii)

- Treatment creates growing heterogeneity.
- Tails separate over time.
- Diagnostic should call for QTT.

The goal is correct classification: stay with ATT when it is enough, and flag QTT when the distribution changes.

Monte Carlo DGP

Untreated outcome:

$$Y_{it}(0) = a_i + \theta_t + \varepsilon_{it}, \quad \theta_t = \frac{t-1}{T-1} + W_t.$$

Pure location shift:

$$Y_{it} = Y_{it}(0) + \beta D_{it}.$$

Growing heterogeneity:

$$Y_{it} = Y_{it}(0) + \beta_{it} D_{it}, \quad \beta_{it} = \lambda_i t^\gamma + v_{it} + \beta.$$

What varies

$N \in \{100, 300, 500, 1000\}$, $T \in \{20, 50, 100\}$, $\beta \in \{0, 0.50\}$, and $\gamma \in \{0.40, 0.50\}$.

Simulation result: ATT is enough

In the pure location-shift design:

- case (i) classification is high,
- performance improves as the post-treatment time dimension grows.

Paper's interpretation

With $T = 100$, case (i) classification is above about 0.90 across reported sample sizes and both average-effect designs.

Takeaway

Short panels are harder, but the false-positive rate falls as T grows.

Simulation result: QTT is needed

In the growing-heterogeneity design with $\gamma = 0.50$:

- treated dispersion rejection is close to one in most designs,
- case (ii)/(iii) classification approaches one with larger N or longer T ,
- lower- and upper-tail statistics carry most of the power,
- the median rejects less often because the design expands tails more than the center.

Takeaway

Weaker divergence, $\gamma = 0.40$, gives lower short-panel power but the same pattern.

Application: Seguro Popular

Seguro Popular expanded non-contributory public health insurance in Mexico.

Economic mechanism

If health insurance becomes available outside formal employment, workers and firms may have weaker incentives to remain formal.

Outcome

Municipal small-firm employment.

Why this application fits

The average effect is close to zero, but lower-tail municipalities may still experience distributional movement.

Empirical workflow in the application

- ① Test pre-treatment control and treated stability.
- ② Test all-period control benchmark stability.
- ③ Check whether sieving is needed.
- ④ Test pre-treatment treated-control quantile comparability.
- ⑤ Test post-treatment treated divergence.
- ⑥ Test post-treatment cross-quantile separation.
- ⑦ Combine with ATT condition.

Seguro Popular: benchmark checks

Sieve results

- Pre-treatment combined treated/control units: 1392 kept, 0 dropped.
- All-period control units: 389 kept, 0 dropped.

Pre-treatment checks

Precondition tests pass at quantiles 0.10, 0.25, 0.50, 0.75, and 0.90.

Interpretation

The post-treatment diagnostic is not being driven by an obviously unstable benchmark.

Seguro Popular: post-treatment diagnostics

Object	<i>t</i> -stat	Decision
Control dispersion $\hat{K}_{N,t}$	-4.77	Non-divergent
Treated dispersion $\hat{K}_{I,t}$	3.16	Diverges
Cross-distance $\hat{K}_{N,t}^{\Delta(0.10)}$	4.97	Diverges
Cross-distance $\hat{K}_{N,t}^{\Delta(0.25)}$	4.12	Diverges
Cross-distance $\hat{K}_{N,t}^{\Delta(0.50)}$	-5.05	Non-divergent
Cross-distance $\hat{K}_{N,t}^{\Delta(0.75)}$	-5.89	Non-divergent
Cross-distance $\hat{K}_{N,t}^{\Delta(0.90)}$	-6.23	Non-divergent

ATT approximately zero; classification: case (ii).

Empirical interpretation

Mean conclusion

Seguro Popular appears to have little average effect on municipal small-firm employment.

Distributional diagnostic

Controls are stable, treated outcomes diverge, and lower-tail quantiles separate from the benchmark.

Classification

Case (ii): no average effect, but lower-tail separation that ATT cannot summarize.

The next step is signed QTT estimation at the flagged lower quantiles.

How to report the diagnostic

Applied papers should report:

- ① control-group weak and distributional non-divergence tests,
- ② treated-group weak and distributional non-divergence tests before treatment,
- ③ whether sieving was used and how many units were retained,
- ④ pre-treatment treated-control quantile-distance tests,
- ⑤ the ATT path,
- ⑥ post-treatment treated-side divergence tests,
- ⑦ cross-quantile separation tests and flagged quantiles.

Then state the implication: ATT enough, common shift, QTT needed, or design warning.

What the diagnostic can and cannot say

It can say

- whether controls are stable,
- whether sieving is needed,
- whether quantile separation appears,
- which quantiles deserve attention.

It cannot say

- the sign of QTT,
- the magnitude of QTT,

Bottom line

The diagnostic points the researcher to the right next empirical step.

Chapter 2: main takeaways

- 1 ATT is often useful, but it is not automatically sufficient.
- 2 Existing estimators provide ways to estimate QTT once assumptions are accepted.
- 3 This paper adds a diagnostic before that step.
- 4 This paper adds a pre-test to check if QTT is required in addition to ATT.
- 5 In the Seguro Popular application, the diagnostic finds lower-tail separation despite near-zero ATT.

Final message

Estimate QTT in addition to ATT when the data show quantile separation.

Chapter 3: why look beyond immigration enforcement?

- County sanctuary policies restrict local cooperation with federal immigration detainers.
- The directly affected people are not the same population as traditional Medicare beneficiaries.
- Households, care workers, providers, and local institutions may transmit the policy shock to older residents.

Research question

Does a verified reduction in the local jail-to-ICE pipeline change Medicare spending or healthcare use in adopting counties?

Chapter 3: possible spillover paths

- ① **Households:** reduced enforcement pressure may change caregiving, mobility, and trust in local institutions.
- ② **Care labor:** workers and providers may change where and how care is supplied.
- ③ **Healthcare use:** previously unmet needs, delayed presentation, or provider behavior may change admissions and spending.

What the design can establish

County-level outcomes can document reduced-form spillovers and test observable implications. They do not identify which household or provider pathway caused them.

Chapter 3: policy and data

Policy variation

- 296 large counties with substantial Hispanic populations.
- 148 adopters and 148 never-adopters, 2007–2024.
- 126 of 148 treated counties adopt in 2013 or 2014.

Linked outcomes

- ICE detainer administrative records for implementation.
- County Medicare spending and utilization.
- QCEW, CBP, and ACS for care-workforce checks.

The estimand pertains to this selected county sample.

Two linked samples answer different questions

Sample	Coverage	Purpose
ICE detainer pipeline	113 counties in 31 states, 2007–2017	Check whether local cooperation changes after adoption.
Medicare county panel	296 selected counties, 2007–2024	Estimate spending and use among Medicare beneficiaries.

- The enforcement and Medicare estimates do not describe the same individuals.
- The first stage validates policy implementation; it is not a mediation estimate.

Chapter 3: empirical design

- ① Define each county's first effective sanctuary-policy year.
- ② Fit the untreated county-year outcome path using not-yet-treated and never-treated observations, county effects, and year effects.
- ③ Compare treated outcomes with that imputed path at each event time.
- ④ Cluster inference at the state level and inspect six pretreatment coefficients.

Identification concern

The design requires untreated outcomes for adopting counties to follow the imputed path. Adoption concentrated in 2013–2014 limits cohort-specific comparisons.

Chapter 3: estimand and comparison

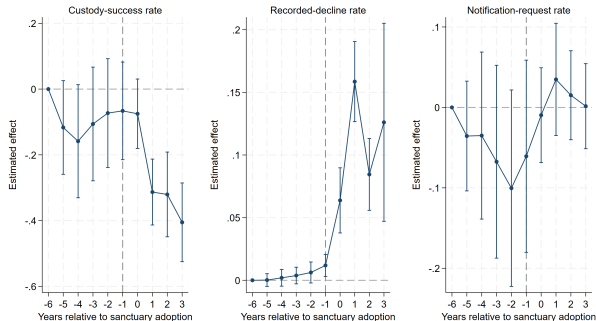
Let G_c be county c 's adoption year and $h = t - G_c$.

$$\hat{\tau}_h = \text{Average}_{c: G_c < \infty, t = G_c + h} \left(Y_{ct} - \hat{Y}_{ct}(0) \right).$$

- Estimate $\hat{Y}_{ct}(0)$ from never-treated and not-yet-treated observations using county and year effects.
- The baseline averages treated *counties*, not beneficiaries, at each supported horizon.
- Long horizons use fewer and earlier adopters; changing support matters for interpretation.

Sanctuary adoption changes the ICE detainer pipeline

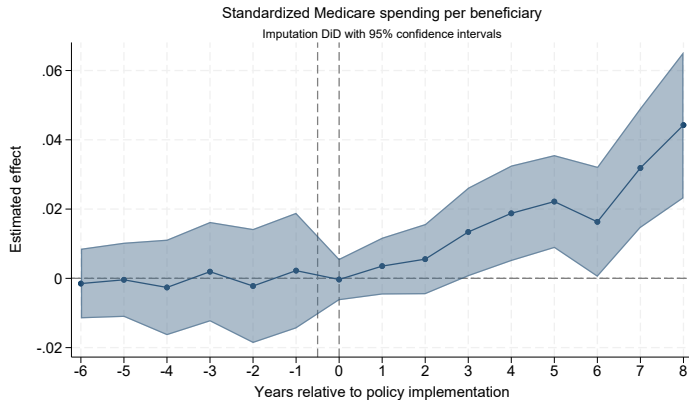
Conditional enforcement margins



- Custody-success rate: -27.8 percentage points, averaged over event years 0–3.
- Recorded-decline rate: $+10.8$ points.
- Notification-request rate: no detectable change.

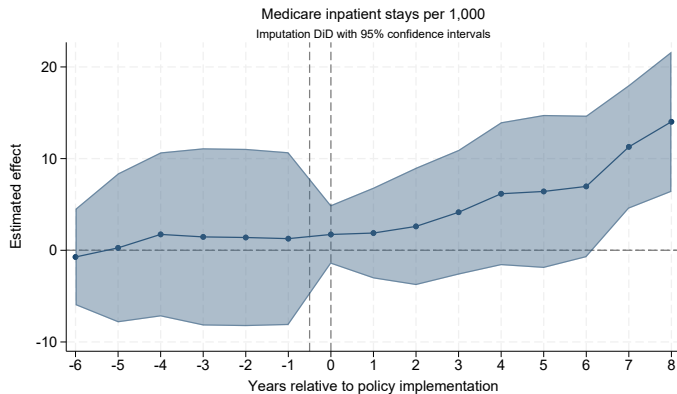
The first stage documents implementation, not a Medicare mechanism.

Medicare event study: standardized spending



Year 5: +2.24%; year 8: +4.52%. Six pretreatment leads: joint $p = 0.183$.

Medicare event study: inpatient stays



Year 8: +14.0 stays per 1,000 beneficiaries; six-lead joint $p = 0.771$.

Medicare spending rises with a delay

Event time	Year 0	Year 5	Year 8
Standardized spending per beneficiary	-0.03%	+2.24%	+4.52%
Inpatient stays per 1,000	+1.73	+6.42	+14.02

- Spending has six jointly insignificant pretreatment coefficients ($p = 0.183$).
- The rise is gradual: the spending estimate is near zero at adoption and positive from year three onward.
- Year-eight spending is a long-horizon comparison supported by fewer treated counties.

Spending robustness: timing and California

Specification	Year 5	Year 8
Baseline	+2.24%	+4.52%
Shift July–December policies forward	+2.30%	+4.91%
Shift October–December forward	+2.44%	+4.97%
Exclude California	+1.66%	+2.69%

- Delayed spending survives alternative annual treatment timing.
- California changes the magnitude materially; the year-eight exclusion estimate sits near the significance boundary.

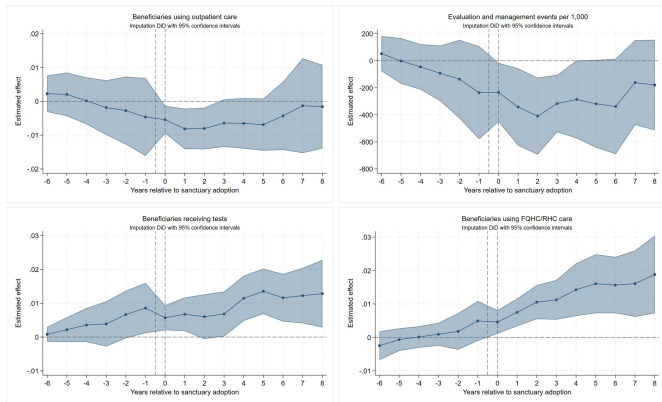
Where does the additional inpatient use appear?

Year-eight outcome	Estimated effect
Beneficiaries with inpatient use	+0.53 percentage points
Covered inpatient days per 1,000	+125.8
Covered days per stay	+0.263 days

- These three outcomes have clean joint pretreatment tests and survive within-family false-discovery-rate adjustment.
- Emergency visits, ambulance events, and readmissions do not increase detectably.

Outpatient event studies complicate the access story

Medicaid expansion and health care access



Outpatient participation and E&M events fall early; testing rises. FQHC/RHC and imaging trends begin before policy adoption.

Mechanism checks do not settle the cause

- QCEW suggests more direct-care employment, but County Business Patterns does not replicate the pattern.
- Public ACS occupation data do not show a robust increase in noncitizen aides or aides per older resident.
- Outpatient participation and evaluation-and-management events fall early; testing rises. FQHC and imaging patterns have problematic pretrends.

Interpretation

The evidence locates a delayed inpatient-utilization response. It does not distinguish improved access, accumulated health needs, provider behavior, coding, or composition.

Care-workforce evidence does not replicate across sources

Source	Result
QCEW industry records	Positive direct-care employment and establishment response.
County Business Patterns	Matched industry measures do not reproduce the QCEW pattern.
Public ACS occupations	Core-aide count rises 0.088 log points, but $q = 0.172$ across ten outcomes; no robust noncitizen-aide or per-older-resident increase.

The workforce pathway remains a hypothesis, not an established mediator.

Who is observed in the Medicare outcomes?

- Standardized spending is reported per traditional Medicare fee-for-service beneficiary.
- Medicare Advantage enrollment and beneficiary risk composition can change the observed fee-for-service population.
- Some composition measures show pretreatment movement, so they cannot be cleanly interpreted as policy responses.
- Changes in the denominator remain an alternative explanation for part of the spending pattern.

Scope

The unweighted design estimates an effect for the average treated county in the selected sample, not a national per-beneficiary fiscal effect.

Cohort-specific comparisons are less reassuring

- The 2013 and 2014 cohorts both have positive long-run pooled spending estimates, but separate six-lead tests reject for each.
- Comparability-adjusted 2014 estimates retain the delayed positive pattern; the year-eight estimate is imprecise and depends on weight treatment.
- The 2013 weighted risk set has weak overlap and rejects pretreatment balance; excluding California reverses its year-eight estimate.

Interpretation

The pooled event study is the main reduced-form result. Cohort checks limit the strength of a causal and nationally generalizable claim.

What limits the sanctuary-policy claim?

- ① Adoption is concentrated in two years; cohort-specific spending pretrend tests reject.
- ② The spending estimate shrinks when California is excluded: 2.69% at year eight, near the significance boundary.
- ③ Traditional Medicare fee-for-service composition may change with Medicare Advantage enrollment or beneficiary characteristics.
- ④ The policy first stage and Medicare response are separate reduced-form findings, not an identified mediation pathway.

The result is a selected-sample spillover, not a national budget estimate.

Chapter 3: current conclusion

- Sanctuary adoption measurably reduces local jail-to-ICE custody conversion.
- Standardized Medicare spending and inpatient use rise several years later in the study counties.
- Multiple data sources weaken a simple care-workforce explanation.
- The welfare meaning and causal pathway remain open.

Contribution

A local immigration-enforcement institution may have consequences for a federal health program serving a different population.

Chapter 4: proposed question

Research question

Does county spending from opioid settlements reduce overdose mortality, and does the answer depend on when, where, and how funds are spent?

- Proposed treatment: actual county settlement expenditures, separated from announced awards.
- Proposed outcome: overdose mortality, with attention to cause coding and small-county suppression.
- Candidate pathways: treatment access, harm reduction, and prevention programs.

Chapter 4: define exposure before estimating effects

- An announced award is not the same as money received or spent.
- Construct county-year measures of actual expenditure per resident and cumulative spending.
- Separate treatment, harm reduction, prevention, and other uses where records permit.
- Allow delayed effects: treatment capacity and mortality may respond on different schedules.

Target

An effect of observed settlement-funded spending on county overdose mortality, for counties with credible spending and mortality measurement.

Chapter 4: design questions before estimation

- ① Can spending dates, amounts, categories, and geographic recipients be measured consistently?
- ② What is the appropriate comparison for counties with different exposure and pre-existing opioid trends?
- ③ How will cross-county service use and drug-market spillovers affect interpretation?
- ④ Are mortality changes plausible at the available follow-up horizon?

Status: design and data construction, with no effect estimate yet.

Chapter 4: proposed empirical sequence

- ① Audit spending records, program locations, and cause-specific mortality coverage.
- ② Plot spending adoption and pretreatment mortality trends before choosing comparison counties.
- ③ Estimate dynamic effects of first substantial spending, with state-year shocks and county differences accounted for where support allows.
- ④ Examine spending categories and neighboring-county exposure; report sensitivity to lag and small-count rules.

The fourth chapter is a design proposal. No mortality effect is estimated yet.

- ① Chapter 1 maps the assumptions behind distributional policy effects.
- ② Chapter 2 develops a diagnostic for control-distribution stability and whether ATT masks quantile separation.
- ③ Chapter 3 measures an indirect health-system response to local immigration enforcement policy.
- ④ Chapter 4 will test whether settlement spending translates into lower overdose mortality.

Common theme: align the estimand, comparison group, and policy mechanism.

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